



UNIVERSITY OF
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Institute of
Computing for
Climate Science

University of
Kent



Programming
Languages and Systems
for Science laboratory

Verifying Climate Models

Chinmayi Baramashetru

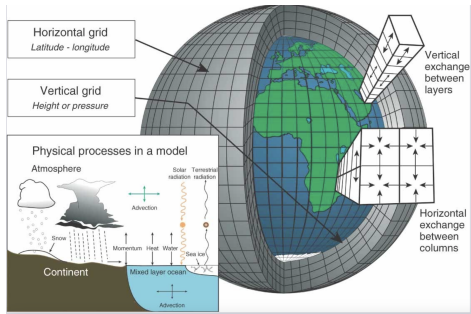
University of Kent

DART Workshop, 2026

Joint work with **Dr. Dominic Orchard**

Climate Models & the gap

Global Circulation Models (GCMs)



Source: Edwards (2011)

- Simulate Earth's climate from physical laws, numerics, and data.
- Implemented as large **Fortran** codes on supercomputers.
- Composed of interacting **submodels**, atmosphere, ocean, land, sea ice etc.
 - Different grid, resolution, time step

A climate model is not one program. It is many programs, glued together.

Software Architecture: the Coupler

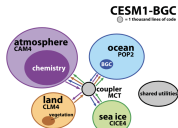


Figure 1. Architecture diagram for CESM1-BGC.

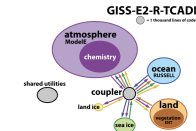


Figure 3. Architecture diagram for GISS-E2-R-TCAD1.

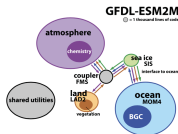


Figure 2. Architecture diagram for GFDL-ESM2M.

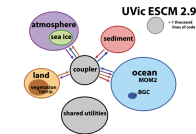


Figure 4. Architecture diagram for UVic ESCM 2.9.

The **coupler** mediates everything between submodels:

- field exchange, interpolation, grid conversion, time synchronisation.
- critical for reproducibility, scalability, correctness.
- small errors propagate and distorts whole result.

Source: Alexander & Easterbrook (2015)

Terminology I will use

Validation

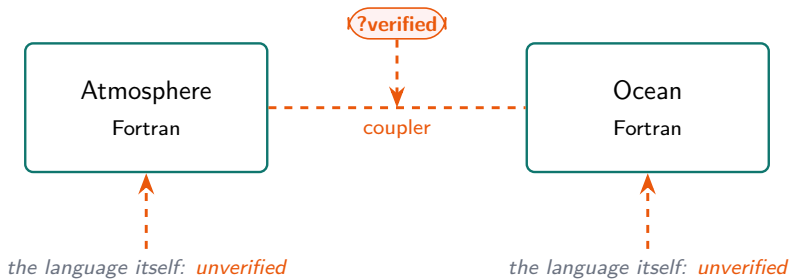
Did we implement the
right equations?

Verification

Did we implement the
equations *right*?

D. Orchard, “Verification in climate science” (NCAR, 2023)

Two Level Issues

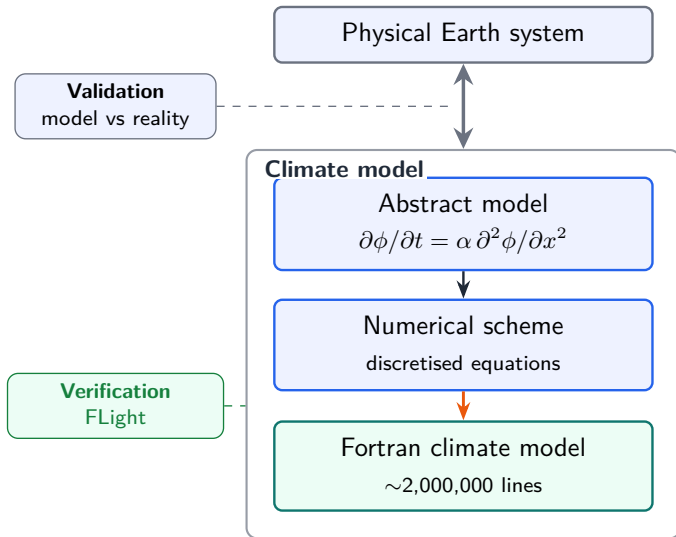


Techniques like testing, deductive verification, proof assistants, model checking verifies **one program in isolation**.

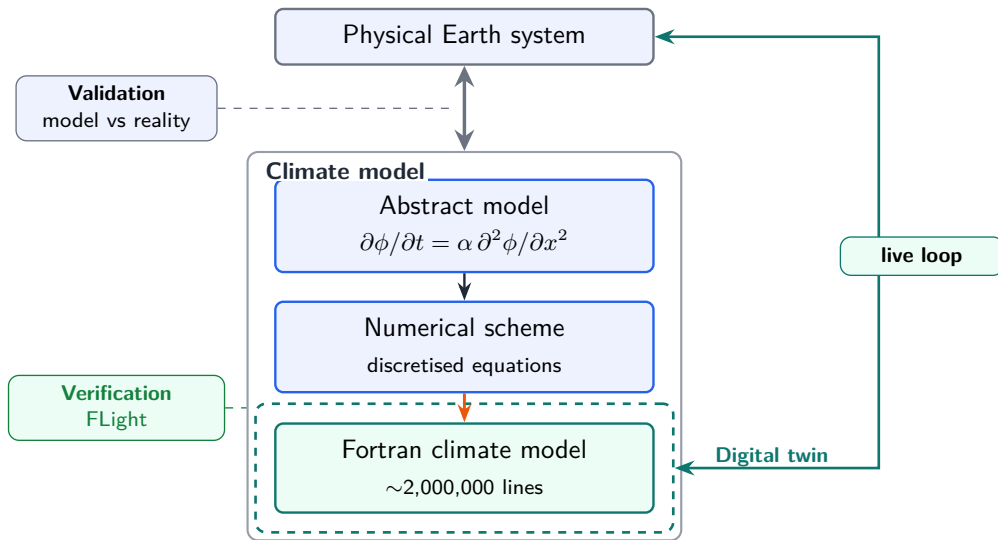
Climate Digital Twins

Physical Earth system

Climate Digital Twins



Climate Digital Twins



Destination Earth: The Climate Change Adaptation Digital Twin

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Model experiment description paper | Highlight paper | 

The Destination Earth digital twin for climate change adaptation

Francisco J. Doblas-Reyes , Jenni Kontkanen, Irina Sandu, Mario Acosta, Mohammed Hussam Al Turjman, Ivan Alsina-Ferrer, Miguel Andrés-Martínez, Costanza Anardi, Leo Arriola, Marvin Axxess, Marc Battle Martin, Peter Bauer, Tobias Becker, Daniel Beltrán, Sebastian Beyer, Hendryk Bockelmann, Pierre-Antoine Bretonnière, Sébastien Cabaniols, Silvia Caprioli, Miguel Castrillo, Aparna Chandrasekar, Suvarchal Cheedela, Victor Correal, Emanuele Danovaro, Paolo Davini, Jussi Enkovaara, Claudia Frauen, Barbara Früh, Aina Gaya Ávila, Paolo Ghinassi, Rohit Ghosh, Supriyo Ghosh, Iker González, Katherine Grayson, Matthew Griffith, Ioan Hadad, Christopher Haine, Carl Hartick, Utz-Uwe Haus, Shane Hearne, Heikki Järvinen, Bernat Jiménez, Amal John, Marlin Juchem, Thomas Jung, Jessica Kegel, Matthias Kelbling, Kai Keller, Bruno Kinoshita, Theresa Kiszler, Daniel Klocke, Lukas Kluff, Nikolay Koldunov, Tobias Kölling, Joonas Kolstela, Luis Kornbluh, Sergey Kosukhin, Aleksander Lacima-Nadolnik, Jeisson Javier Leal Rojas, Jonni Lehtiranta, Tuomas Lunttila, Anna Luoma, Pekka Manninen, Alexey Medvedev, Sebastian Milins Ali Mohammed, Sebastian Müller, Devaraju Naryanappa, Natalia Nazanova, Sami Niemelä, Bimochan Niraula, Henrik Nortamc, Aleksii Nummellin, Matteo Nurisso, Pablo Ortega, Stella Paronuzzi, Xabier Pedruzo-Bagazgoitia, Charles Pelletier, Carlos Peña, Suraj Polade, Himansu Kesari Pradhan, Rommel Quintanilla, Tiago Quintino, Thomas Rackow, Jouni Räisänen, Maqsood Mubarak Rajput, René Redler, Balthasar Reuter, Nuno Rocha Monteiro, Francisc Roura-Adserias, Silva Ruppert, Susan Sayed, Reiner Schnur, Tanvi Sharma, Dmitry Sidorenko, Outi Sievi-Korte, Albert Soret, Christian Steger, Björn Stevens, Jan Streffing, Jaleena Sunny, Luiggi Tenorio, Stephan Thober, Ulf Tigerstedt, Oriol Tinto, Juha Tonttila, Heikki Tuomenvirta, Lauri Tuppi, Ginka Van Thielien, Emanuele Vitali, Jost von Hardenberg, Ingo Wagner, Nils Wedi, Jan Wehner, Sven Willner, Xavier Yepes-Arbós, Florian Ziemer, and Janos Zimmermann

Abstract

The development of a digital twin of the Earth system is at the core of the ambitious EU initiative *Destination Earth* (DestinE). Implemented by ECMAF, ESA and EUMETSAT, under the leadership of the Directorate General Communications Networks, Content and Technology (DG-CONCT).

Article Assets Peer review Metrics Related art

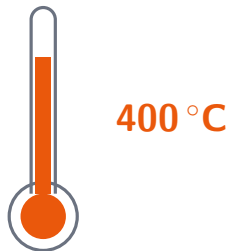
14 Apr



Verifying the Interface

Interface Contracts

- Unit / dimension mismatch across the boundary
- Grid & resolution mismatch
- Time-step misalignment
- Out-of-range fields at the receiver



Interface Contracts

- Verifying full models is extremely complex; couplers are **critical and shared**.
- Complex, **loosely specified** orchestration layers - hard to keep correct, reproducible, maintainable.
- Widely reused: **130+** couplers across major CMIP¹ projects.
- Verify one coupler $\Rightarrow$ benefit **many** models at once.
- Behaves like software components: **contracts**, modular reasoning.

¹Coupled Model Intercomparison Project.

Major coupler frameworks



CPL6 / CPL7



Earth System Modelling Framework



OASIS



Model Coupling Toolkit

Common functional modules across couplers

- Couplers differ in design (standalone vs. embedded, timing, interpolation)
- but share recurring tasks.
- These recurring tasks become verifiable modules and design principles.

Configuration and Initialisation Module

Field Register and Metadata Module

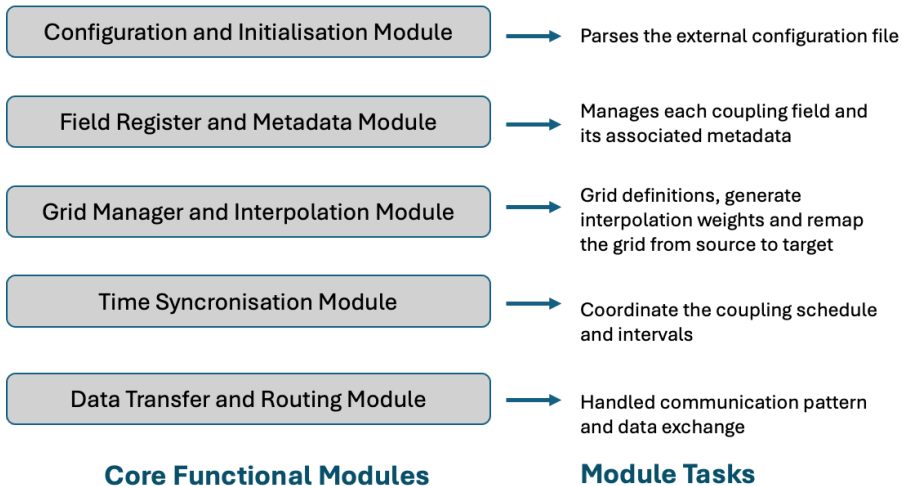
Grid Manager and Interpolation Module

Time Synchronisation Module

Data Transfer and Routing Module

Core Functional Modules

From modules to contracts

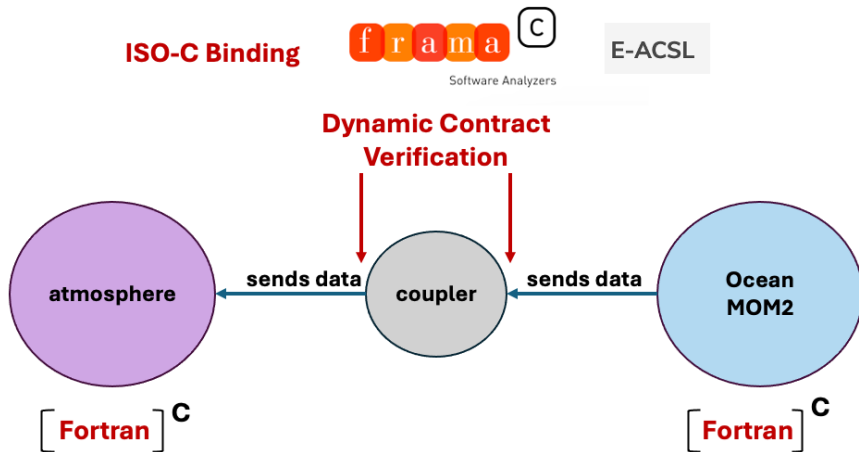


Example: time-synchronisation module

```
/*@  
  requires all_components_reached_time(t);  
  requires clocks_compatible;  
  
  ensures  data_exchanged_at(t);  
  ensures  no_duplicate_or_missed_exchange;  
  
  invariant time_monotonic:  
    time_next >= time_current;  
*/  
void sync_step(double t);
```

- **Pre:** components aligned, clocks consistent.
- **Post:** data exchanged exactly once per step t .
- **Invariant:** simulation time never moves backwards.
- Expressible in ACSL enables **contract verification**.

Dynamic contract verification (Frama-C)



Runtime assertions check data flow and consistency [at the coupler boundary](#).

Verifying the language: FLight

Fortran dominance

Aspect	- Full migration to Python costs \$15 million - Complete rewrites or migrations are risky: bit-for-bit fidelity, validation, and legacy code complexity.		
Current Fortran Usage in Weather/Climat			
WMO Compliance Requirements			
Performance Comparison (Fortran vs Pyt			
Migration Complexity	High - Years of effort, specialized expertise required	Very High	Very High
Skills Availability	Critical shortage - 1,050 Fortran jobs vs 83,391 Python jobs	Very High	High
Training Requirements	Months to years for scientists to learn new languages	High	Medium
Legacy Code Maintenance	Expensive - Millions of lines of tested, validated code	Very High	Very High
Interoperability Solutions	Fortran-Python interfaces available but add overhead	Medium	Medium

Source: Uy, J. (2024)



- Fortran lacks the modern **formal verification tools** (Frama-C, KeY) that C and Java enjoy.
- The gap blocks deductive reasoning and machine-checked proofs of what the code computes.

*Cam*Fort

Verification Techniques Used in Climate Models

- Testing is widely used.
- Other verification efforts (e.g., Interactive theorem provers)

A Coq Formalization of Taylor Models and Power Series for Solving Ordinary Differential Equations

Sewon Park 

Graduate School of Informatics, Kyoto University, Japan

Holger Thies 

Graduate School of Human and Environmental Studies, Kyoto University, Japan

A Verified ODE Solver and the Lorenz Attractor

Fabian Immler¹ 

Wave Equation Numerical Resolution: A Comprehensive Mechanized Proof of a C Program

Sylvie Boldo · François Clément ·
Jean-Christophe Filliâtre · Micaela Mayero ·
Guillaume Melquiond · Pierre Weis

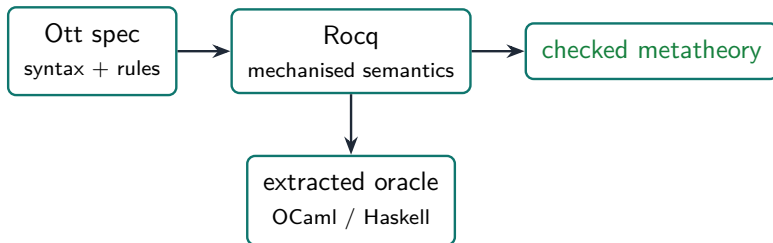
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Abstract A rigorous numerical algorithm, formally verified with Isabelle, certifies the computations that Tucker used to prove chaos for the Lorenz attractor. The formalization is based on a formalization of a diverse variety of mathematics. Formalized mathematics include ordinary differential equations and Poincaré maps. The algorithm includes low level approximation schemes based on Runge–Kutta methods and interval arithmetic. On a high level, reachability analysis is guided by static hybrid

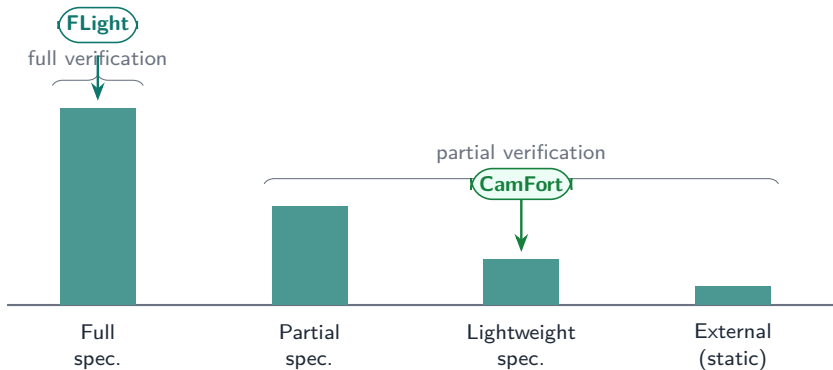
FLight: a verifiable subset of Fortran 90

- A subset of Fortran 90 with a real, [machine-checked operational semantics](#).
- Mechanised in the [Rocq](#) proof assistant, generated via [Ott](#)
- [Executable](#) as a reference oracle via Rocq extraction (OCaml/Haskell)
- *one* definition documents, runs, *and* proves

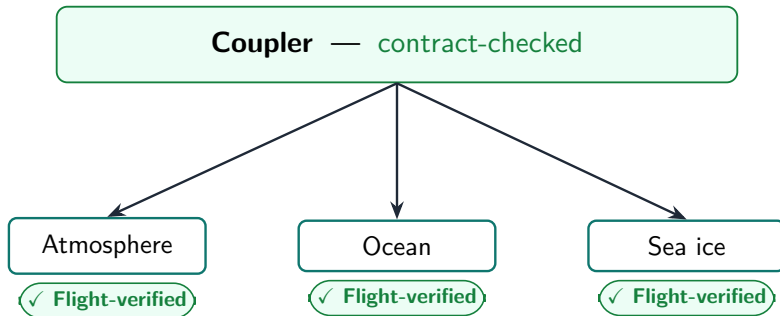


joint work with P. Sewell

Verification Efforts



The whole system



Current Status

- FLight is an **early subset** of Fortran 90 (no COMMON, EQUIVALENCE, derived types, coarrays yet).
- The contracts are **partial**, targeting the highest-risk seams.
- The extracted oracle is for **litmus-sized** grids.

Thank you

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